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NOTE

# UNIVERSAL TILINGS AND UNIVERSAL (0, 1)-MATRICES

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A periodic regular tiling of the plane by black and white squares is $k$-universal if it contains all possible $k \times k$ blocks of black and white tiles. There is a $4 \times 4$ periodic tiling that is 2-universal; this paper looks for the smallest 3-universal tiling and obtains a $64 \times 32$ periodic tiling that is 3-universal.

Related to this is the following: a $(0, 1)$-matrix is $k$-universal if every possible $k \times k$ $(0, 1)$-matrix occurs as a submatrix. It is proved that, for $k$ even, there is a $k2^{k/2}$ by $k2^{k/2}$ matrix that is $k$-universal and, for $k$ odd, there is a $(3k+1)2^{(k-3)/2}$ by $(3k-1)2^{(k-3)/2}$ matrix that is $k$-universal.

## 1. Introduction

Let us consider regular tilings of the plane by unit squares coloured black and white. Such a tiling is called *periodic* if the design can be obtained, by translations through multiples of $m$ units vertically and $n$ units horizontally, from a fundamental $m \times n$ block of tiles. We may speak of this as an $m \times n$ periodic tiling. The fundamental block can be represented by a $(0, 1)$-matrix, that is, an $m \times n$ matrix with entries 0 and 1, corresponding to the black and white tiles.

Such a tiling is called *k-universal* if it contains all the possible $k \times k$ blocks of black and white tiles. It is always possible to find a $k$-universal tiling by simply taking a fundamental block consisting of the $2^{k^2}$ different $k \times k$ blocks stacked together, but one is naturally interested in finding a smaller $k$-universal tiling. In the case $k = 2$, stacking together to form a square fundamental block would give an $8 \times 8$ periodic tiling but there is in fact a $4 \times 4$ periodic tiling that is 2-universal. This is the tiling, found for example in [2], with fundamental block given by the matrix

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 \end{bmatrix}.$$

In this paper we shall consider the case $k = 3$ and obtain what we conjecture to be the smallest 3-universal tiling. This is the tiling with the $64 \times 32$ fundamental block given in Fig. 1.

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| | 1 | 3 | 7 | 3 | 2 | 3 | 8 | 3 | 1 | 4 | 7 | 4 | 2 | 4 | 8 | 4 | 1 | 5 | 7 | 5 | 2 | 5 | 8 | 5 | 1 | 6 | 7 | 6 | 2 | 6 | 8 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 7 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 8 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 9 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 9 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 2 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 3 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 7 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 4 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 8 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 7 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 2 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 8 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 3 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 9 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 4 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 9 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |

Fig. 1.

Fig. 1. (*Continued*)

| | 1 | 3 | 7 | 3 | 2 | 3 | 8 | 3 | 1 | 4 | 7 | 4 | 2 | 4 | 8 | 4 | 1 | 5 | 7 | 5 | 2 | 5 | 8 | 5 | 1 | 6 | 7 | 6 | 2 | 6 | 8 | 6 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 10 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 3 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 7 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 4 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |
| 8 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 6 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 |

Fig. 1.

As a first step in constructing this 3-universal tiling, we shall also consider $k$-universal (0, 1)-matrices. An $m \times n$ matrix is *$k$-universal* if every possible $k \times k$ (0, 1)-matrix occurs as a submatrix, in the sense that it can be obtained by selecting some $k$ rows and some $k$ columns, these rows (and columns) not necessary adjacent or in increasing order. Let

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}.$$

We claim that the matrix $A$ is 3-universal. For example, the $3 \times 3$ matrix

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

is the submatrix obtained by selecting the 2nd, 7th and 3rd rows and the 5th, 1st and 6th columns of $A$. So, although the fundamental block of a periodic tiling is a (0, 1)-matrix, the two senses of $k$-universal are different: The fundamental block being a $k$-universal matrix does not make a tiling $k$-universal because, in the case of a tiling, each possible $k \times k$ (0, 1)-matrix must be found as a block within the tiling, and not simply as a submatrix by a suitable selection of rows and columns.

We shall describe a construction that gives the following result:

**Theorem.** *For $k$ even, there is a $k2^{k/2}$ by $k2^{k/2}$ matrix that is $k$-universal; for $k$ odd, there is a $(3k+1)2^{(k-3)/2}$ by $(3k-1)2^{(k-3)/2}$ matrix that is $k$-universal.*

The proof of this makes up Section 3 below. For $k=3$, this gives the $10 \times 8$ 3-universal matrix $A$ shown above, and from this is derived the $64 \times 32$ fundamental block of a 3-universal tiling.

## 2. A 3-universal tiling

By stacking together all the $2^9$ different $3 \times 3$ blocks of black and white tiles, one cannot obtain a 3-universal tiling with a square fundamental block, but one obtains, for example, a $3 \cdot 2^5 \times 3 \cdot 2^4$ periodic tiling this way. One would like to find a smaller 3-universal tiling than this and the computer printout shown in Fig. 1 gives a $64 \times 32$ periodic tiling that is 3-universal. The next two sections describe how this was found, but there are three facts about it worth noting first:

(1) This 3-universal tiling has in fact just 10 different rows and 8 different columns, and the numbers or labels appearing down the left-hand side and along the top indicate this property, identical rows (or columns) having the same label. These are derived from the 10 rows and 8 columns of the 3-universal matrix given above;

(2) The number of $3 \times 3$ blocks in this tiling is $2^{11}$, so some of the $2^9$ different possible $3 \times 3$ blocks appear more than once. However, suppose the entry in the $i$th row and $j$th column is $a_{ij}$ and let $A_{ij}$ be the $3 \times 3$ block with $a_{ij}$ in its top left-hand corner. Then the number of $3 \times 3$ blocks $A_{ij}$ with $i$ and $j$ *both odd* is $2^9$ and these in fact are all different. So every possible $3 \times 3$ block occurs exactly once as one of these;

(3) A fabric that is periodic is also given by a fundamental $m \times n$ block of black and white squares and the question of $k$-universal fabrics has been raised (for example [2, p. 159]). In the terminology of that paper, Fig. 1 gives a strongly 3-universal fabric except that it is not isonemal. By considering row-sums and column-sums one can check, using [1], that it does form a fabric that hangs together.

## 3. The construction of a $k$-universal (0, 1)-matrix

There is a natural correspondence which associates with any (0, 1)-matrix a bipartite tournament. The problem of finding a $k$-universal (0, 1)-matrix is thus the same as the similar problem for bipartite tournaments and the following is based on the method for tournaments in general given in [3, pp. 49–50].

Let $M$ be any $k \times k$ (0, 1)-matrix and let $u_1, u_2, \ldots, u_k$ be the rows of $M$ and $v_1, \ldots, v_k$ the columns of $M$. It is convenient to use the notation $e(u_i, v_j)$ for the entry in row $u_i$ and column $v_j$. Now let

$$U_i = \{v_j\colon e(u_i, v_j) = 0\},$$

$$V_j = \{u_i\colon e(u_i, v_j) = 1\}.$$

Construct a matrix $A$ with rows $r_{i,S}$ in one-to-one correspondence with the ordered pairs $(i, S)$, where $S \subset U_i$ and columns $c_{j,T}$ in one-to-one correspondence with the ordered pairs $(j, T)$, where $T \subset V_j$. The entry in row $r_{i,S}$ and column $c_{j,T}$ of this matrix will be denoted by $e(r_{i,S}, c_{j,T})$ and is defined by these rules:

$$\text{If } e(u_i, v_j) = 1, \text{ then } e(r_{i,S}, c_{j,T}) = \begin{cases} 1 & \text{if } u_i \in T \\ 0 & \text{if } u_i \notin T. \end{cases}$$

$$\text{If } e(u_i, v_j) = 0, \text{ then } e(r_{i,S}, c_{j,T}) = \begin{cases} 0 & \text{if } v_j \in S \\ 1 & \text{if } v_j \notin S. \end{cases}$$

We now show that $A$ is $k$-universal. Let $B$ be any $k \times k$ (0, 1)-matrix with rows $x_1, \ldots, x_k$ and columns $y_1, \ldots, y_k$. The entry in row $x_i$ and column $y_j$ is denoted by $e(x_i, y_j)$. Define

$$f(x_i) = r_{i,S(i)}, \quad \text{where } S(i) = \{v_j : e(u_i, v_j) = 0 \text{ and } e(x_i, y_j) = 0\},$$

$$f(y_j) = c_{j,T(j)}, \quad \text{where } T(j) = \{u_i\colon e(u_i, v_j) = 1 \text{ and } e(x_i, y_j) = 1\}.$$

Then we claim that the submatrix of $A$ defined by the rows $f(x_i)$ and the columns $f(y_j)$ is equal to the $k \times k$ matrix $B$. This is because if $e(u_i, y_j) = 0$, then

$$e(x_i, y_j) = 0 \Rightarrow v_j \in S(i) \Rightarrow e(r_{i,S(i)}, c_{j,T(j)}) = 0 \Rightarrow e(f(x_i), f(y_j)) = 0,$$

$$e(x_i, y_j) = 1 \Rightarrow v_j \notin S(i) \Rightarrow e(r_{i,S(i)}, c_{j,T(j)}) = 1 \Rightarrow e(f(x_i), f(y_j)) = 1$$

and if $e(u_i, v_j) = 1$, then

$$e(x_i, y_j) = 1 \Rightarrow u_i \in T(j) \Rightarrow e(r_{i,S(i)}, c_{j,T(j)}) = 1 \Rightarrow e(f(x_i), f(y_j)) = 1,$$

$$e(x_i, y_j) = 0 \Rightarrow u_i \notin T(j) \Rightarrow e(r_{i,S(i)}, c_{j,T(j)}) = 0 \Rightarrow e(f(x_i), f(y_j)) = 0.$$

The number of rows in $A$ is equal to $2^{|U_1|} + \cdots + 2^{|U_k|}$ and the number of columns is equal to $2^{|V_1|} + \cdots + 2^{|V_k|}$. Now $|V_j|$ is equal to the column-sum of the column $v_j$ and $|U_i|$ is equal to $k$ minus the row-sum of the row $u_i$ of the arbitrary matrix $M$. To minimise the number of rows and columns of $A$, we choose a matrix $M$ with row-sums and column-sums as nearly equal as possible. Thus if $k$ is even we can choose $M$ with all its row-sums and column-sums equal to $\frac{1}{2}k$. Then the number of rows and columns in $A$ is $k2^{k/2}$.

If $k$ is odd, we choose $M$ so that $\frac{1}{2}(k+1)$ of the row-sums and of the column-sums are equal to $\frac{1}{2}(k-1)$ and $\frac{1}{2}(k-1)$ of the row-sums and of the column-sums are equal to $\frac{1}{2}(k+1)$. Then the number of rows in $A$ works out to be $(3k+1)2^{(k-3)/2}$ and the number of columns $(3k-1)2^{(k-3)/2}$. In the case $k = 3$,

this gives a $10 \times 8$ matrix, and the 3-universal matrix $A$ given in the introduction was obtained by taking $M$ to be

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

The method gives a $16 \times 16$ matrix that is 4-universal and a $32 \times 28$ matrix that is 5-universal.

## 4. The final step

We return now to the problem of constructing a 3-universal tiling. We will not give all the details of this work but it can be seen that Fig. 1 was built up by using the 10 rows and 8 columns of the 3-universal matrix $A$ found by the method of Section 3 and given in the introduction. What we had to ensure was that every *essentially different* choice of 3 rows from the 10 rows of the matrix $A$ appeared as 3 consecutive rows in the required order in the larger matrix and that the similar property for columns held. This was done partially by hand and checked by computer.

## References

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